

SILENT FOLDS: PROJECTIVE POLES AS A TRAINING-FREE FAILURE FLAG IN HOMOGRAPHY-BASED RETINAL REGISTRATION

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ABSTRACT

Homographies remain the default transform in feature-based retinal registration, yet every homography has a pole line on which its denominator vanishes; where that line crosses the fundus the warp folds, while the matrix still looks valid. Auditing 625 FIRE and COph100 pairs registered with XFeat, SuperPoint/LightGlue, and the released retina-specific SuperRetina, we find that all 100 observed image crossings are registration failures, 25 of them not flagged by the estimator. The pole clearance ρ , a unit-free distance from four denominator evaluations, is a training-free quality-control (QC) score: 167 of the 168 estimates within one image diagonal of a pole fail. Fused with RANSAC inlier ratio, it reaches an area under the ROC curve of 0.85–0.86, at most 0.04 below a learned detector (95% bound), and stays at or above 0.75 across failure endpoints where either score alone drops to 0.47–0.56. Poles can also destabilize learned QC: unbounded curvature features reversed Platt calibration in 20 detector folds, all holding one patient’s poles; guarding, log-transforming, or dropping the feature prevents this, and only a guard keeps the pole as an explicit flag.

Index Terms— retinal image registration, quality control, homography, failure detection, calibration

1. INTRODUCTION

Registration of fundus photographs underlies longitudinal monitoring and screening, for example of retinopathy of prematurity in infants [1, 2]. A silently failed registration corrupts every downstream change measurement, so at scale failures must be caught automatically. Feature-based retinal methods still estimate a homography: SuperRetina fits one to learned keypoints [3], and SuperJunction uses one as its first stage [4]. Registration quality control (QC), in turn, predicts failure from appearance, match, and transformation features such as Jacobians and bending energy [5, 6, 7].

A homography is a rational map. Where its projective denominator changes sign it sends image points through infinity and reverses orientation, a *fold*: in a fundus overlay, the moving image is torn along the pole line and the part beyond it is mirrored onto the far side. Close to that line its derivatives are unbounded. RANSAC scores reprojection consensus, and the returned matrix is finite. Robust-estimation frameworks test sample and model degeneracy [8], and Moisan et al. reject “wild homographies” by enforcing orientation preservation at the four sample points [9]; both constrain the sample, not the support the warp is applied to.

Contributions. (1) An audit of three modern retinal pipelines on FIRE and COph100 [10, 1], including the released SuperRetina: every observed pole crossing is a failure, a quarter of them silent. (2) Pole clearance as a QC score. Poles and the rectangle test are not new; we give exact tests for the rectangular, circular, and convex

supports a warp is applied to, normalize the pole distance into a continuous, unit-free clearance, and measure it against inlier ratio, projective baselines, and a learned detector. (3) A calibration caveat: unbounded homography-derived features can reverse Platt calibration of learned QC, and a guard must contain the feature’s support.

2. METHOD

Pole and folding. A homography H maps (x, y) to

$$u = \frac{h_{11}x + h_{12}y + h_{13}}{d}, \quad v = \frac{h_{21}x + h_{22}y + h_{23}}{d}, \quad (1)$$

with denominator $d(x, y) = h_{31}x + h_{32}y + h_{33}$ and Jacobian determinant $\det J(x, y) = \det H / d(x, y)^3$ [11]. Hence $\det J$ changes sign exactly where d does: the pole line $d = 0$ is the fold, and $|\det J|$, like every higher derivative, diverges as $d \rightarrow 0$.

Exact support tests. Because d is affine, its extrema on an axis-aligned rectangle lie at the four corners c_k , so the pole crosses the closed rectangle if and only if

$$\min_k d(c_k) \leq 0 \leq \max_k d(c_k). \quad (2)$$

Remark: this is the algebraic form of the common check that the projected corners form a convex quadrilateral with the source orientation: a corner triangle’s image orientation is $\text{sign}(\det H) \text{sign}(d_i d_j d_k)$, so the two agree unless a corner lies on the pole. Unlike the projected check, Eq. (2) extends to any convex support, such as a fundus mask’s convex hull, by evaluating d at its vertices (for a non-convex mask this is conservative). For a circular field of view (FOV) with center q and radius r , with $a = (h_{31}, h_{32})$, the condition is $|a^\top q + h_{33}| \leq r \|a\|_2$; we use the image’s inscribed circle as the FOV proxy. H (Frobenius-normalized, so $h_{33} = 0$ is allowed) maps moving to fixed image and is tested on the moving support; QC scores use H only, and H^{-1} on the fixed image is reported separately.

Clearance. For a non-crossing rectangle with diagonal L ,

$$\rho(H) = \min_k |d(c_k)| / (L \|a\|_2) \quad (3)$$

is the distance from the image to the pole line in image diagonals ($\rho = 1$: the pole is one diagonal away); $\rho = 0$ for a crossing and $\rho = \infty$ for an affine map. It is invariant to matrix scale and image resolution.

Uses. (i) *QC score:* $s = -\log_{10} \max(\rho, 10^{-3})$, with ρ capped at 10^3 . The *fused* score averages the empirical percentiles of s and of the negated RANSAC inlier ratio, each percentile map estimated on non-test groups only (cross-fitted over the five group folds of Sec. 3), so a new case is scored alone; no label is used. (ii) *Feature guard:* a derivative feature is set missing when the pole crosses the support on which it is computed. (iii) *Support-constrained RANSAC:*

Case 1: pole clips a corner, misses the FOV

Case 2: pole crosses the FOV

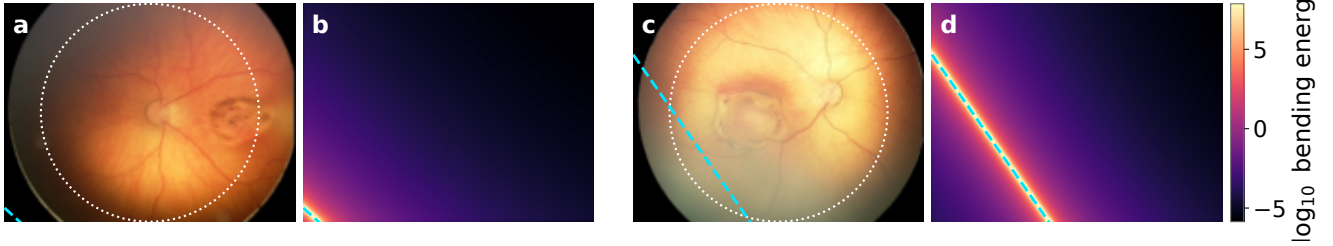


Fig. 1. Two failed Coph100 XFeat-H registrations (patient 055). (a,c) Moving fundus images; (b,d) log bending-energy density of the estimated warp on one shared scale. Cyan: pole line $d = 0$; dotted: inscribed-FOV proxy. In case 1 the pole clips the rectangle but misses the FOV and the fundus mask; in case 2 it crosses the FOV along a 294.9-pixel chord. Images: RIDIRP (CC0); annotations: Coph100 (CC BY 4.0) [2, 1].

Table 1. Pole audit on 625 retinal pairs. Failed / flagged: returned transforms that fail / that the criterion flags (2/2: both flagged estimates failed). AUROCs are pooled and group-weighted over all attempted pairs, with 95% group-bootstrap intervals; inlier ratio is a pre-specified training-free baseline (for SuperRetina, its LMedS inlier rate); fused: Sec. 2; detector: median raw AUROC of the guarded Base detector over 21 assignments, trained for the generic pipelines only. Constrained RANSAC (rectangle): failures made successes / silent failures made explicit / explicit failures made silent; for SuperRetina, the study RANSAC re-fits its stage-1 matches. SIFT-H is a weak-matching stress test.

Pipeline	Returned	Failed	Failed / flagged		$\rho \leq 1$	AUROC				Constr. RANSAC
			Rect. cross.	FOV cross.		$-\log \rho$	Inlier ratio	Fused	Detector	
XFeat-H	625	342	2/2	1/1	20/21	.778 (.71-.84)	.848 (.80-.90)	.855 (.80-.90)	.833	1/0/0
SPLG-H	618	278	10/10	6/6	40/40	.823 (.77-.87)	.782 (.71-.85)	.850 (.79-.90)	.838	0/4/0
SuperRetina	568	277	88/88	81/81	107/107	.852 (.80-.90)	.811 (.75-.87)	.859 (.80-.91)	–	29/2/72
SIFT-H	521	519	375/375	187/187	412/412	–	–	–	–	0/0/104

minimal-sample hypotheses whose pole crosses the support are discarded before consensus scoring, on the same random stream; if the consensus refit crosses, the best admissible hypothesis is returned, and if none exists the registration fails explicitly.

3. EXPERIMENTAL SETUP

Data and endpoint. We use 134 FIRE [10] and 491 Coph100 [1] pairs, grouped into 42 image-connected FIRE components (patient identifiers are unavailable) and 56 Coph100 patients. A registration fails if it returns no transform (an *explicit* failure) or its mean landmark error exceeds 0.005 of the fixed-image diagonal in original coordinates (a *silent* failure): 20.6 pixels on FIRE (2912²), within its standard 0–25-pixel success curve, and 4 pixels on Coph100 (640 × 480).

Pipelines. The generic (not retina-specific) pipelines XFeat-H, SPLG-H, and SIFT-H match XFeat [12], SuperPoint/LightGlue [13], or SIFT [14] correspondences at a 1,024-pixel long edge (Coph100 upsampled 1.6×) and share one seeded RANSAC with a normalized DLT (3-px threshold, 10⁴ iterations, consensus refit). SuperRetina is run as released (least-median-of-squares, LMedS, with its two-pass refinement) at original resolution [3]; it reproduces the published FIRE success-curve AUCs of the S/P/A (large-overlap/small-overlap/anatomical-change) categories within 0.011.

Learned detector. Per pipeline, an L2-regularized logistic regression combines image, match, inverse-consistency, geometric (including bending energy $\mathcal{B}$ on a 32 × 32 lattice, our curvature feature), cross-pipeline, and structure features (Base); a second arm adds re-sampling and perturbation stability (+Stab). Features are standardized after training-median imputation with one missingness indicator each, and $C \in \{0.1, 1, 10\}$ is chosen by grouped inner cross-validation.

Platt scaling [15], an unregularized logistic fit of the raw score, uses a disjoint calibration fold; a negative slope reverses the ranking. With five group folds (test f , calibration $f+1$, training the rest), we refit everything under 21 seeded group-to-fold assignments and pool test predictions with equal total weight per patient group. Assignment spread reuses cases and is not a confidence interval; intervals are 95% group bootstraps (2,000 resamples of patients or components) on the original assignment. No-output pairs receive the most-failing value of every training-free score.

4. RESULTS

4.1. Pole crossings are failures, often silent

All 12 crossings by the generic matchers are failures (Table 1). SuperRetina with its own LMedS returns 88 crossing homographies, all failures; 75 have no LMedS inliers, which its evaluation counts as failed, but 13 would pass silently. Testing H^{-1} on the fixed image adds 4 SPLG-H crossings (4 failed); none of the estimates is a global reflection. The projected-corner check flags the same crossings, as it must; the 32 × 32 lattice det J test costs 256 times more and misses 3/375 SIFT-H crossings. Under weak matching, invalid geometry is common: 72% of SIFT-H homographies cross the image. Crossings alone have low recall (12 of 620 generic-matcher failures); the clearance extends them: within one diagonal, 60 of 61 generic-matcher and all 107 SuperRetina estimates fail (Fig. 2b). All crossings fail up to 0.01 of the diagonal and 99/100 at 0.02. That crossings fold is a mathematical fact; that they all fail is an observation on these data, and the events are not independent: 7 of the 12 generic-matcher crossings come from one patient (055) of 4, and SuperRetina’s span 38 patients with at most 10 each. No FIRE pair crosses, against 1.2%

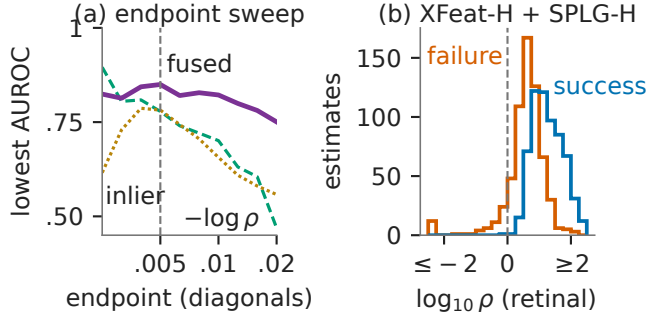


Fig. 2. (a) Lowest AUROC over the three pipelines as the failure endpoint varies (dashed: study endpoint). (b) Clearance of all returned XFeat-H and SPLG-H estimates; ≤ -2 holds the 12 crossings. Dashed: $\rho = 1$, left of which 60/61 estimates fail.

(generic) and 18% (SuperRetina) of COph100 pairs; adult and infant cohorts also differ in camera and resolution.

4.2. Clearance is a training-free QC score

$-\log \rho$ is weaker than inlier ratio on XFeat-H and stronger on SPLG-H and SuperRetina (Table 1). The two are complementary: the fused score exceeds inlier ratio on SPLG-H and SuperRetina (paired 95% intervals of the AUROC difference [0.03, 0.11] and [0.02, 0.08]) but not on XFeat-H ($[-0.03, 0.04]$), and is at most 0.04 AUROC below the learned detector (paired intervals $[-0.01, 0.07]$, $[-0.04, 0.06]$). ρ is not carried by the crossings: on estimates with $\rho > 1$ it reaches .772/.808/.700 (XFeat-H/SPLG-H/SuperRetina) versus .842/.794/.667 for inlier ratio, and no AUROC moves by more than 0.03 on returned transforms only. Any measure of how projective H is ranks like ρ : the pole distance from the image center and the corner deviation from H 's best affine fit agree within ± 0.010 (paired intervals), while the condition number (0.65–0.76) and warped-area scale (0.68/0.70 on XFeat-H/SPLG-H) are weaker. The support thus matters for the crossing decision and the unit threshold, not for ranking. Flagging $\rho \leq \tau$ with τ chosen on training groups for $\geq 95\%$ precision keeps 92%/94% held-out precision at 20%/35% recall (median $\tau = 2.34/2.93$ diagonals). Across endpoints (Fig. 2a), ρ improves on SPLG-H and SuperRetina but falls on XFeat-H, whose near-pole estimates are often only moderately wrong, and inlier ratio degrades on SPLG-H; the fused score's lowest AUROC is 0.75, versus 0.47 and 0.56.

4.3. Poles can destabilize learned calibration

Raw homography-derived curvature can reverse a detector's Platt calibration; here every reversal traces to one patient's poles. In one held-out fold (XFeat-H, +Stab, original assignment), the raw detector ranks failures well (AUROC 0.927), but its Platt slope is -4.3×10^{-5} , compressing all 83 test probabilities into [0.57961, 0.57970] and reversing them (0.073). The calibration fold held the two patient-055 poles of Fig. 1, whose bending energies (8.65×10^9 , 1.05×10^9) lie far outside the training range; deleting both restores the fit, deleting either does not. All 20 reversals over 21 assignments have patient-055 poles in the calibration fold, so they repeat one mechanism rather than adding evidence. The cause is geometric: moving a synthetic pole from $\rho = 1$ to 10^{-3} outside a 640×480 image raises lattice bending energy from 5.53 to 2.13×10^7 .

Raw derivative features combined with Platt scaling are therefore fragile. Guarding, log-transforming, or dropping them prevents ev-

Table 2. Calibration over 21 assignments. Reversals (105 folds per cell) for the Base/+Stab arms; XFeat-H Base calibrated AUROC and Brier score as median (worst assignment). Curvature and guard supports are crossed; “none” is unguarded. The last rows agree to three decimals but differ per case. SPLG-H worst Brier: .175 raw, .167 guarded. [†]One fit with an optimizer warning.

Curv. supp.	Guard supp.	Reversals		XFeat-H Base	
		XFeat-H	SPLG-H	AUROC	Brier
Rect.	none	13 [†] /6	1/0	.787	.180 (.206)
Rect.	FOV	13 [†] /6	1/0	.788	.180 (.207)
FOV	none	11/6	0/0	.802	.178 (.207)
Rect.	Rect.	0/0	0/0	.828	.169 (.179)
FOV/FOV, log, drop		0/0	0/0	.828	.169 (.181)

ery reversal and lowers the worst-assignment Brier score (Table 2); clipping and quantile transforms behave alike, and isotonic calibration [16] prevents reversal but raises the median NLL to 0.97 (0.51 guarded). A guard must contain the feature's support: sampling curvature only inside the FOV leaves 17 reversals, because the pole can cross the FOV itself, and an FOV guard on rectangle-sampled curvature removes none, because a pole can clip the rectangle while missing the FOV (Fig. 1a); over-guarding is harmless. Once the poles are neutralized, curvature adds little signal. The guard's value is diagnostic: it keeps curvature where it is valid and records the pole as an explicit flag, which marked a failure in all 100 observed crossings, whereas removal and monotone transforms hide it.

4.4. Support-constrained RANSAC

Moisan's sample-point test [9] still returns 13 crossings; the support constraint returns none, but fold-free is not error-free: it rescues none of the 12 generic-matcher pole cases and, under weak matching, turns explicit failures into wrong estimates (Table 1), so its output still needs QC.

5. DISCUSSION AND CONCLUSION

A finite homography is not necessarily a valid warp on the fundus: if its pole intersects the support, the map is singular there and folds. Released retinal pipelines return such homographies, some silently, and in our experiments every one was a registration failure. A few denominator evaluations detect them exactly on rectangular, circular, or convex supports, and the clearance turns the same quantity into a training-free QC score that adds to inlier ratio; fused, it comes within 0.04 AUROC of a learned detector without training. Learned QC should guard, transform, or drop homography-derived derivative features; a support-containing guard additionally keeps the pole as an explicit flag. Constraining RANSAC removes folds but not errors. We recommend these checks as low-cost safeguards in homography-based retinal registration: reject pole-crossing estimates, report ρ alongside the inlier ratio, and guard derivative features on their evaluation support. The study is limited to two public retinal datasets, generic-matcher poles from four patients (SuperRetina's span 38), FIRE groups that may share unobserved patients, and homographies. Quadratic retinal models [17, 18] and eye-model registration [19] have no projective denominator; homographies persist in learned pipelines because they are estimable from four points, and they are applied even to infant retinopathy-of-prematurity imaging, where all poles of the three modern pipelines in our data occur.

6. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made available in open access by FIRE [10], RIDIRP [2], and COph100 [1]. FIRE reports written informed consent; RIDIRP reports consent and approval by the Ethical Committee of University Hospital Ostrava (544/2018). Ethical approval was not required for this secondary analysis, which used only the publicly released, de-identified data; no new participant data were collected.

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